Prabhavit Gupta

Founding AI Voice Engineer: Real-Time Speech Agents, Full-Duplex Audio & LLM Systems

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Summary

Founding AI Voice and Agent Systems Engineer with 3 years of commercial experience designing, building, and deploying real-time conversational agents, voice-to-voice pipelines, and autonomous LLM architectures. Hands-on experience developing low-latency streaming audio solutions (WebSockets, WebRTC, OpenAI Realtime API, Whisper STT, ElevenLabs/Cartesia TTS), full-duplex turn-taking, interruption handling, and affective persona steering. Deep technical foundation in stateful context and memory persistence (LangGraph, Redis), dynamic RAG knowledge retrieval, safety guardrails, and automated evaluation harnesses. Proven track record taking AI products from zero to production at startup velocity. Permanent EU resident, available immediately.

Experience

FlowSpace.co • *Lead AI Engineer & Conversational Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production real-time conversational agents serving 4,500+ active users with sub-1.8s end-to-end response latency.
- Implemented full-duplex streaming audio pipelines utilizing WebSockets, Server-Sent Events (SSE), and OpenAI Realtime API with voice activity detection (VAD) and graceful interruption recovery.
- Built persistent multi-session memory, context compaction, and affective persona steering using LangGraph, Redis, and vector search (pgvector).
- Developed robust safety guardrails and moderation layers ensuring empathetic, safe, and hallucination-free interactions across sensitive conversational domains.
- Designed automated evaluation harnesses (LangSmith, PyTest) measuring dialogue coherence, turn-taking latency, audio quality, and token/inference costs.
- Owned backend microservices from prototype to high-availability deployment using Python (FastAPI), TypeScript, Docker, and AWS.
- *Tools: Python 3, TypeScript, WebSockets, OpenAI Realtime API, Whisper, LangGraph, Redis, pgvector, FastAPI, Docker, AWS*

Urlink • *AI Systems & Integration Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated semantic extraction and classification pipelines connecting external partner APIs with centralized database stores.
- Implemented structured schema validation and context retrieval layers in Python, ensuring resilient data processing.
- Built RESTful connectors integrating autonomous agent outputs into production client workflows.
- *Tools: Python, REST APIs, LangChain, Embeddings, PostgreSQL, Supabase, Git, Linux*

Independent Technical Consultant • *Voice & AI Solutions Engineer*
Sep 2023 - Mar 2024 • Venice, Italy

- Designed custom conversational voicebots and conversational assistants for European clients integrating speech-to-text, LLM reasoning, and text-to-speech synthesis.
- Built custom function calling and tool execution wrappers to enable conversational agents to perform external tasks during live dialogue.
- Authored technical architecture runbooks, prompt engineering guidelines, and deployment specifications for client engineering teams.
- *Tools: Python, OpenAI API, LangChain, REST APIs, WebSockets, Linux, Git*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Built automated webhook consumers and data integration workers in Python for SaaS client onboarding flows.
- *Tools: Python, REST APIs, Sendgrid*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization across the EU and for worldwide contracts. No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working intermediate).
- **Availability:** Available immediately (0 notice period) for full-time remote roles.

Core Skills

- **Voice AI & Streaming Audio:** STT/TTS Pipelines (Whisper, ElevenLabs, Cartesia), OpenAI Realtime API, WebSockets, WebRTC, VAD, Interruption Handling.
- **Agent Architecture & Memory:** LangGraph, LangChain, Multi-Agent Systems, Stateful Dialogue, Memory Systems, Context Compaction, Tool Calling.
- **Safety & Evaluation:** Guardrails, Content Moderation, Automated Dialogue Evals (LangSmith), Latency & Cost Optimization.
- **Full-Stack & Backend:** Python 3, TypeScript, FastAPI, Node.js, REST APIs, Redis, pgvector, Docker, AWS, Git.
- **Startup Mindset:** Rapid Prototyping, Zero-to-One Product Ownership, Autonomous Execution, High-Velocity Shipping.